PROMPT ENGINEERING CASE STUDY

Systematic Prompt Engineering & Auditing for Genomic Data Extraction: A RARS1 Case Study

A progressive study in prompt design — Version 0 through Version 5

SOURCE ARTICLE (IDENTICAL ACROSS ALL VERSIONS)

Mutations in the RARS1 gene, which encodes the cytoplasmic arginyl-tRNA synthetase, are a well-documented cause of severe neurodevelopmental disorders, primarily characterized by hypomyelination, spastic tetraparesis, and nystagmus. In this study, we investigated three pediatric patients presenting with progressive psychomotor regression. Using Western blot analysis on patient-derived skin fibroblasts, we observed a significant decrease in RARS1 protein expression. To model the disease pathway, we utilized CRISPR-Cas9 gene editing to knock out RARS1 in human embryonic kidney cells, which revealed a severe impairment in cytoplasmic translation and downstream mitochondrial respiration deficits. These findings confirm that RARS1 mutations destabilize the enzyme, leading to systemic translation failure.

VERSION 0

Baseline (General Summary)

PROMPT

```
"Summarize this article about the RARS1 gene for me."
```

OUTPUT

RARS1 encodes cytoplasmic arginyl-tRNA synthetase, and its mutations are known to cause severe neurodevelopmental disorders marked by hypomyelination, spastic tetraparesis, and nystagmus. This study looked at three pediatric patients with progressive psychomotor regression.

Key findings:

- **Reduced protein levels**: Western blot analysis of patient-derived skin fibroblasts showed significantly decreased RARS1 protein expression.
- **Functional modeling**: CRISPR-Cas9 knockout of RARS1 in human embryonic kidney cells caused severe impairment of cytoplasmic translation, along with downstream mitochondrial respiration deficits.
- **Conclusion**: The mutations appear to destabilize the RARS1 enzyme, leading to systemic translation failure that likely drives the disease phenotype.

In short, the paper links RARS1 mutations to reduced enzyme stability, which impairs protein synthesis and has knock-on effects on mitochondrial function, offering a mechanistic explanation for the neurodevelopmental symptoms seen in patients.

EVALUATION NOTES

Provided a general narrative overview. Useful as a starting point but not optimized for extracting specific, categorized data (findings vs. symptoms vs. methods were blended together).

VERSION 1

Adding a Goal / Task Specification

PROMPT

```
"Summarize this article about the RARS1 gene for me. My goal is to extract only the actual findings, clinical
symptoms of mutations, and the specific research methods used, rather than just getting a general overview."
```

OUTPUT

Clinical Symptoms of RARS1 Mutations

- Hypomyelination
- Spastic tetraparesis
- Nystagmus
- Progressive psychomotor regression (observed in the three patients in this study)

Research Methods Used

- Western blot analysis on patient-derived skin fibroblasts (to measure RARS1 protein expression)
- CRISPR-Cas9 gene editing to knock out RARS1 in human embryonic kidney cells (to model the disease pathway)

Actual Findings

- Patient fibroblasts showed significantly decreased RARS1 protein expression compared to normal
- RARS1 knockout cells showed severe impairment in cytoplasmic translation
- Knockout cells also showed downstream mitochondrial respiration deficits
- Conclusion: RARS1 mutations destabilize the enzyme, causing systemic translation failure

EVALUATION NOTES

Adding a defined goal successfully separated the content into distinct categories (symptoms, methods, findings), improving scannability and relevance over the baseline. Still presented as prose-style bullet lists rather than a strictly formatted structure.

VERSION 2

Adding a Defined Audience

PROMPT

```
"Summarize this article about the RARS1 gene for me. My goal is to extract only the actual findings, clinical
symptoms of mutations, and the specific research methods used, rather than just getting a general overview. Your
audience is 10-year-old elementary school children who are visiting a science museum."
```

OUTPUT

What Happens When the RARS1 Gene Has a Problem?

Our bodies have thousands of genes that act like instruction manuals. The RARS1 gene's job is to help build a special tool that helps cells make proteins (proteins are like tiny machines that keep our bodies running). When RARS1 doesn't work right, kids can have some tough health problems:

- Trouble with the coating that protects nerve wires in the brain (like insulation around an electrical wire)
- Stiff, hard-to-control arms and legs
- Eyes that move back and forth on their own without meaning to
- Losing skills they'd already learned, like sitting or talking

How Did Scientists Study This?

Scientists used some cool tools to investigate:

- They took tiny skin samples from three kids who had these symptoms and looked at the proteins inside their cells using a special test
- They used a tool called CRISPR (kind of like molecular scissors!) to remove the RARS1 gene from cells in a lab, so they could see what happens when it's missing

What Did They Discover?

- The kids' cells had way less of the RARS1 tool than normal
- When scientists removed RARS1 from lab cells, those cells had a hard time building proteins at all
- This also caused problems with the cell's "power plants" (called mitochondria), which make energy for the cell
- So the scientists figured out that when RARS1 is broken, cells can't build proteins properly, which causes big problems throughout the body

EVALUATION NOTES

The audience constraint successfully shifted tone and simplified language for a 10-year-old reader. However, this came at the cost of precision: technical terminology (e.g., "arginyl-tRNA synthetase," "cytoplasmic translation," "fibroblasts") was stripped out or replaced with metaphors. This made the output worse for professional/clinical use, prompting a pivot away from a child audience in the next version.

VERSION 3

Adding Strict Constraints (Professional, Concise, No Metaphors)

PROMPT

```
"Summarize this article about the RARS1 gene for me. My goal is to extract only the actual findings, clinical
symptoms of mutations, and the specific research methods used, rather than just getting a general overview.
Constraints: The entire summary must be under 150 words total, use professional scientific terminology, and must not
contain any metaphors or analogies."
```

OUTPUT

Clinical presentation

RARS1 mutations are associated with severe neurodevelopmental disorders characterized by hypomyelination, spastic tetraparesis, and nystagmus. Three pediatric patients presented with progressive psychomotor regression.

Methods

Western blot analysis was performed on patient-derived skin fibroblasts to quantify RARS1 protein expression. CRISPR-Cas9 gene editing was used to generate RARS1 knockout human embryonic kidney cells as a disease model.

Findings

Patient fibroblasts exhibited significantly decreased RARS1 protein expression. RARS1 knockout cells demonstrated severe impairment of cytoplasmic translation and downstream mitochondrial respiration deficits. These results indicate that RARS1 mutations destabilize the enzyme, resulting in systemic translation failure.

Word count: 122

EVALUATION NOTES

Reintroducing precise scientific terminology and removing the child-audience constraint restored professional accuracy. The strict word limit (150 words) and no-metaphor rule produced a dense, publication-appropriate summary at 122 words. The only remaining usability issue: the content, while accurate and concise, was still presented as a block of prose, making rapid visual scanning difficult.

VERSION 4

Adding an Output Format (Markdown Table)

PROMPT

```
"Summarize this article about the RARS1 gene for me. My goal is to extract only the actual findings, clinical
symptoms of mutations, and the specific research methods used, rather than just getting a general overview.
Constraints: The entire summary must be under 150 words total, use professional scientific terminology, and must not
contain any metaphors or analogies. Output Format: Organize the output as a Markdown table with exactly three
columns: 'Symptom Profile', 'Biological Cause', and 'Lab Methodology'."
```

OUTPUT

Symptom Profile	Biological Cause	Lab Methodology
Hypomyelination	Decreased RARS1 protein expression destabilizes cytoplasmic arginyl-tRNA synthetase	Western blot analysis on patient-derived skin fibroblasts
Spastic tetraparesis	Impaired cytoplasmic translation due to RARS1 deficiency	CRISPR-Cas9 knockout of RARS1 in human embryonic kidney cells
Nystagmus	Downstream mitochondrial respiration deficits secondary to translation failure	Assessment of mitochondrial respiration in knockout cell model
Progressive psychomotor regression	Systemic translation failure resulting from enzyme destabilization	Comparative protein expression analysis (patient vs. control)

Word count: 79

EVALUATION NOTES

The table layout dramatically improved scannability compared to the prose block in Version 3, while staying well under the word limit (79 words vs. the 150-word cap). This format is well-suited for quick reference or presentation use. However, the row-by-row pairing of symptoms to causes is an organizational simplification — the source article does not actually establish a distinct, individualized causal chain per symptom. This raised the question of how to verify and audit the table's claims against the source, leading to the final version.

VERSION 5

Adding Verification Requirements (Final)

PROMPT

```
"Summarize this article about the RARS1 gene for me. My goal is to extract only the actual findings, clinical
symptoms of mutations, and the specific research methods used, rather than just getting a general overview.
Constraints:
1. The entire summary (the table content) must be under 150 words total.
2. Use professional scientific terminology.
3. Do not contain any metaphors or analogies.
Output Format:
Organize the output as a Markdown table with exactly three columns: 'Symptom Profile', 'Biological Cause', and 'Lab
Methodology'.
Verification Requirements:
Directly below the table, create a section titled 'Verification Sources'. In this section, provide direct, exact
sentence quotes from the source text that verify the claims made in each of the three columns."
```

OUTPUT

Symptom Profile	Biological Cause	Lab Methodology
Hypomyelination	Decreased RARS1 protein expression destabilizes cytoplasmic arginyl-tRNA synthetase	Western blot analysis on patient-derived skin fibroblasts
Spastic tetraparesis	Impaired cytoplasmic translation due to RARS1 deficiency	CRISPR-Cas9 knockout of RARS1 in human embryonic kidney cells
Nystagmus	Downstream mitochondrial respiration deficits secondary to translation failure	Assessment of mitochondrial respiration in knockout cell model
Progressive psychomotor regression	Systemic translation failure resulting from enzyme destabilization	Comparative protein expression analysis (patient vs. control)

VERIFICATION SOURCES

- **Symptom Profile (Hypomyelination, Spastic tetraparesis, Nystagmus):** *"Mutations in the RARS1 gene, which encodes the cytoplasmic arginyl-tRNA synthetase, are a well-documented cause of severe neurodevelopmental disorders, primarily characterized by hypomyelination, spastic tetraparesis, and nystagmus."*
- **Symptom Profile (Progressive psychomotor regression):** *"In this study, we investigated three pediatric patients presenting with progressive psychomotor regression."*
- **Biological Cause (protein destabilization / decreased expression):** *"Using Western blot analysis on patient-derived skin fibroblasts, we observed a significant decrease in RARS1 protein expression."* Also: *"These findings confirm that RARS1 mutations destabilize the enzyme, leading to systemic translation failure."*
- **Biological Cause (impaired translation and mitochondrial deficits):** *"To model the disease pathway, we utilized CRISPR-Cas9 gene editing to knock out RARS1 in human embryonic kidney cells, which revealed a severe impairment in cytoplasmic translation and downstream mitochondrial respiration deficits."*
- **Lab Methodology (Western blot):** *"Using Western blot analysis on patient-derived skin fibroblasts, we observed a significant decrease in RARS1 protein expression."*
- **Lab Methodology (CRISPR-Cas9):** *"To model the disease pathway, we utilized CRISPR-Cas9 gene editing to knock out RARS1 in human embryonic kidney cells."*

Note on verification limits: The source article does not establish a distinct mechanistic link between each individual symptom (e.g., nystagmus specifically) and a distinct biological cause — the causal claims (protein destabilization, translation impairment, mitochondrial deficits) are reported collectively, not symptom-by-symptom. The table's row-by-row pairing is an organizational simplification for scanning purposes, not a claim the source text itself makes at that granularity. This is worth flagging if this table is used for clinical or publication purposes, since the audit trail shows several rows draw on the same underlying sentences.

EVALUATION NOTES

Adding a verification layer transformed the table from an unaudited summary into a traceable, source-linked artifact. Every cell's claim can now be checked against an exact sentence from the original text, which is critical for scientific/clinical trust. This also surfaced an important limitation: the neat one-to-one table structure imposes more granularity than the source text actually supports, since several rows trace back to the same one or two sentences. This is a valuable finding in its own right — it shows that formatting choices (like a table) can visually imply a level of precision that isn't present in the underlying source, and that verification steps are what catch this kind of subtle overreach.

Summary: What the Prompt Ladder Demonstrated

Version	Technique Added	Key Improvement	Key Trade-off / Limitation
0	None (baseline)	Basic coverage of the article	Unstructured, blended categories
1	Goal / task specification	Clear separation into symptoms/methods/findings	Still prose-heavy
2	Audience definition	Simplified, accessible tone	Lost technical precision (bad for professional use)
3	Strict constraints	Restored precision, concise (122 words), no metaphors	Still a scanning-unfriendly text block
4	Output format (table)	Highly scannable, very concise (79 words)	Table implies false granularity/precision
5	Verification requirements	Fully auditable, source-linked, trustworthy	Reveals and documents the table's precision limits

Overall conclusion: Each layer of prompt engineering (goal, audience, constraints, format, verification) solved the specific weakness introduced or exposed by the previous version, illustrating that better prompts are built iteratively rather than in a single pass.